

1. Mastering the translation between tensor language and component expressions

Solving these problems provides you with all key skills for tensor calculations.

Let $\varphi : V \rightarrow V$ be a linear map on a vector space $(V, \oplus, \odot)$ and $e_1, \dots, e_{\dim V}$ a basis for V .

- (a) Prove that a linear map $\varphi : V \rightarrow V$ defines a $(1, 1)$ -tensor

$$T_\varphi : V^* \times V \rightarrow \mathbb{R}, \quad T(\omega, v) := \omega(\phi(v)).$$

- (b) Show that defining the components φ^i_j of the linear map φ as the expansion coefficients for the image $\varphi(e_j)$ of the j -th basis vector e_j in terms of the same basis in the target space, namely

$$\varphi(e_j) = \bigoplus_{i=1} \varphi^i_j \odot e_i,$$

one has

$$\varphi^i_j = (T_\varphi)^i_j,$$

where $(T_\varphi)^i_j$ are the components of the tensor T_φ .

- (c) Does the equality in (b) depend on the choice of basis? Conclude how the components of a linear map transform under a change of basis. How, in contrast, do the components of a bilinear map $B : V \times V \rightarrow \mathbb{R}$ transform under the same change of basis?
- (d) Derive that the components $(\psi \circ \varphi)^i_j$ of the composition $\psi \circ \varphi : V \rightarrow V$ of two linear maps $\varphi : V \rightarrow V$ and $\psi : V \rightarrow V$ are given by

$$(\psi \circ \varphi)^i_j = \sum_m^{\dim V} \psi^i_m \varphi^m_j.$$

Explain how this result gives rise to the well-known multiplication rule for matrices.

- (e) The *dual map* of a linear map $\varphi : V \rightarrow V$ is the linear map

$$\varphi^* : V^* \rightarrow V^*, \quad \sigma \mapsto \sigma(\varphi(\cdot)),$$

where $\sigma(\varphi(\cdot)) : V \rightarrow \mathbb{R}$ is the linear map defined by $v \mapsto \sigma(\varphi(v))$ and is thus indeed an element of the dual space V^* . Show that expanding the image of the dual map on $(V^*, \boxplus, \boxminus)$,

$$\varphi^*(\sigma) = \bigoplus_{j=1}^{\dim V} \left(\sum_{i=1}^{\dim V} \varphi^*_{j,i} \sigma_i \right) \boxminus e^j,$$

identifies the components $\varphi^{*,i}_j = \varphi^i_j$. How is the matrix of components for φ^* related to the one for φ ?

- (f) A linear map $\varphi : V \rightarrow V$ is called *symmetric with respect to an inner product* $g : V \times V \rightarrow \mathbb{R}$ if

$$g(\varphi(v), v') = g(v, \varphi(v')) \quad \text{for all } v, v' \in V.$$

Express this relation entirely in terms of components with respect to some basis and then rewrite it in terms of

$$\varphi_{ij} := \sum_{m=1}^{\dim V} g_{im} \varphi^m_j.$$

Simplify the result as much as possible using the properties of inner products.